Strategy-Space Modeling for Interpretable Quant Research

From Linear Combination to ML-Guided Allocation on GLD

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Starting Point

- The project started from a mathematical question, not a trading rule.
- In class, we studied model selection using polynomial basis functions:

$$1,\ x,\ x^2,\ \ldots,\ x^d$$

- That raised the main idea:

Can trading strategies also be treated as basis functions?

- If a strategy maps a market state into a numerical score, it can be evaluated across dates and placed into a design matrix.

Function Space Analogy

Concept	Polynomial Model	Strategy Model
basis function	$1, x, x^2, \ldots$	strategy signal functions
data point	x_i	market state at date t_i
matrix entry	$\phi_j(x_i)$	strategy score $s_j(t_i)$
model selection	choose degree / penalty	choose strategy / penalty
target	observed y_i	future cumulative return

Core design matrix:

$$A_{ij} = \phi_j(\text{market state at } t_i)$$

Each column is a strategy function evaluated over time. Each row is one trading day.

Four Trading Strategy Families

These are strategies practitioners actually use on price data.

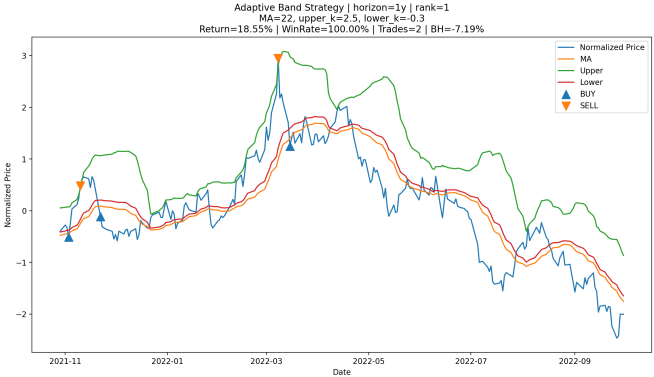
Strategy	Signal Type	Core Idea
Adaptive Band	continuous	price position inside rolling bands
MA Crossover	continuous	short vs. long moving average gap
Adaptive Volatility Band	continuous	volatility regime from H-L-C range
Fear-Greed Candle-Volume	event	large candle + volume spike

Each strategy maps a market state to a number (score).
That number becomes one column of matrix A.

Strategy 1: Adaptive Band

Signal type: continuous

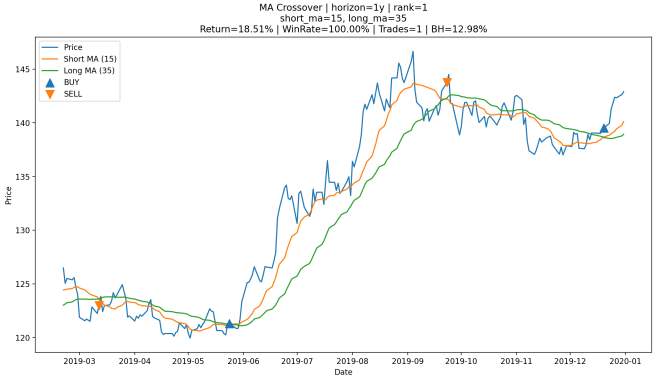
- Price normalized relative to a rolling moving average
- Upper and lower bands placed at scaled thresholds
- Score: how far price sits from the band midpoint
- Buy zone: price near or below lower band
- Sell zone: price near or above upper band



Strategy 2: Moving-Average Crossover

Signal type: continuous

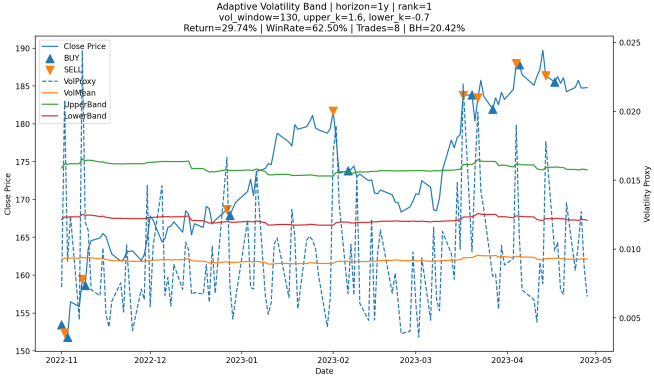
- Short-window MA compared against long-window MA
- Positive spread: uptrend; negative spread: downtrend
- Score captures direction and magnitude of the gap
- Buy: short MA crosses above long MA
- Sell: short MA crosses below long MA



Strategy 3: Adaptive Volatility Band

Signal type: continuous

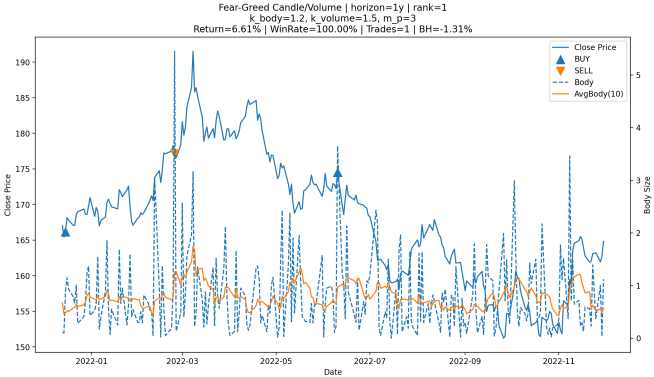
- Volatility proxy from high-low-close range
- Bands placed relative to rolling volatility level
- Score reflects current volatility regime position
- High-vol breakouts and low-vol compressions produce distinct signals
- Complementary to price-band strategies



Strategy 4: Fear-Greed Candle-Volume

Signal type: event-style

- Fires only when candle body size **and** volume spike simultaneously
- Captures extreme sentiment: large directional move confirmed by activity
- Rare by design; most days produce no signal
- Qualitatively different from the three continuous strategies
- Adds a sparse basis vector to matrix A



Turning Parameters into Signals

Each strategy family has its own parameters. After optimization, the best parameters define a concrete signal function for that anchor.

Example: anchor date 2021-12-31

Strategy 1 — Adaptive Band (MA=65, upper_k=0.0, lower_k=0.6)

$$s_{\text{band}}(t) = \frac{P_t - \text{MA}_{65}(t)}{\sigma_{65}(t)}, \quad \text{buy signal when } s_{\text{band}}(t) < 0.6$$

Strategy 2 — MA Crossover (short=60, long=65)

$$s_{\text{ma}}(t) = \text{MA}_{60}(t) - \text{MA}_{65}(t), \quad \text{buy signal when } s_{\text{ma}}(t) > 0$$

- Parameters are re-optimized at every anchor date (monthly)
- Each anchor produces a different function adapted to recent market behavior
- The score value on any day becomes one entry in matrix A

Stage A: What Happens with One Strategy at a Time?

Hold 130 days · 1-month parameter update · rank-1 parameters per anchor

Year	B&H	Adap.Band	MA Cross	Adap.Vol	Fear-Greed
2020	+23.9%	+4.4%	+5.2%	+5.8%	≈0%
2021	−6.2%	+0.3%	−0.2%	+0.9%	≈0%
2022	+0.8%	−0.9%	−3.0%	−2.1%	≈0%
2023	+11.8%	+1.5%	+0.9%	+2.1%	≈0%
2024	+27.0%	+5.9%	+6.3%	+10.6%	≈0%

- Each strategy alone captures only a **fraction** of the market move
- Exposure is low (21–46%): the strategy sits out most days
- Fear-Greed: after removing look-ahead bug, GLD rarely triggers the threshold (≈0% exposure)

Reading the Numbers: Capital Efficiency

Why does our strategy return less than Buy-and-Hold?

Buy-and-hold is **always 100% exposed**. Our strategy sits out when the signal is weak — gross exposure typically ranges from 21% to 78%.

A fairer comparison: **return per unit of capital deployed**.

$$\text{Efficiency} = \frac{\text{Return}}{\text{Avg. Exposure}}$$

2024	Return	Avg. Exposure	Efficiency
Buy-and-Hold	+27.0%	100%	+27%
Stage C, hold 130 days	+16.4%	65%	+25%
Stage C, hold 45 days	+25.2%	78%	**+32%**

- Hold-45 Stage C **exceeds** B&H efficiency in 2024
- The idle capital (22–79%) can be deployed into other assets

Stage B: Combining Strategies — Matrix A

Single strategies each capture a partial signal. **Can combining them do better?**

Define the design matrix A where each column is one strategy score:

$$A = \begin{pmatrix} s_1(t_1) & s_2(t_1) & s_3(t_1) & s_4(t_1) \\ s_1(t_2) & s_2(t_2) & s_3(t_2) & s_4(t_2) \\ \vdots & \vdots & \vdots & \vdots \\ s_1(t_n) & s_2(t_n) & s_3(t_n) & s_4(t_n) \end{pmatrix}$$

Find weights $\mathrm{x} \geq 0$ (long-only) that maximize realized return y:

$$\max_{\mathrm{x} \geq 0,\ 1^\top \mathrm{x} = 1} \quad y(\mathrm{x}) \ = \ \text{portfolio return using combined signal } A\mathrm{x}$$

This is **Objective 1**: a constrained optimization over the strategy space.
Parameters are re-optimized at each anchor date (rolling walk-forward).

Stage B: What Did the Optimizer Actually Find?

Anchor: 2021-12-31 — selection period return vs. buy-and-hold: **+10.2% vs −6.2%**

Long-only weight optimization over the 4 strategy signals:

$$\hat{s}(t) = \underbrace{0.45}_{\text{Adaptive Band}} \cdot s_{\text{band}}(t) + \underbrace{0.00}_{\text{MA Cross}} \cdot s_{\text{ma}}(t) + \underbrace{0.55}_{\text{Adap.Vol}} \cdot s_{\text{vol}}(t) + \underbrace{0.00}_{\text{Fear-Greed}} \cdot s_{\text{fg}}(t)$$

- Two strategies dominate: Adaptive Band (45%) and Adaptive Vol (55%)
- MA Crossover and Fear-Greed received zero weight — optimizer excluded them
- This mirrors Lasso-style selection: only the most informative signals survive
- Standalone returns in selection period: Band=+10.0%, Vol=+10.1%, MA=−5.6%

The combined signal $\hat{s}(t)$ then drives portfolio exposure via the tranche system.

Stage B: Ensemble Results

Hold 130 days · 1-month update · long-only weight re-optimization per anchor

Year	B&H	Stage A (best single)	Stage B (ensemble)	Improvement
2020	+23.9%	+4.4%	+6.4%	↑
2021	−6.2%	+0.3%	+0.3%	∼
2022	+0.8%	−0.9%	−2.3%	↓
2023	+11.8%	+1.6%	+2.0%	↑
2024	+27.0%	+5.9%	+8.0%	↑

- Ensemble beats single strategy in 3/5 years, but margin is small
- Exposure stays low (23–42%): combined signal is still weak
- 2022: ensemble *worse* than best single (−2.3% vs −0.9%) — diversification hurts in whipsaw markets
- **The weights are fixed after optimization — they cannot adapt to what the market is doing now**

What Stage B Cannot Do

Stage B finds the best *historical* weight vector x.

But the market changes. The optimal combination today may differ from last month.

The missing step:

learn how to combine signals to predict **future** return

	Stage B	Stage C
Question	best historical weights?	what predicts future return?
Method	constrained optimization	supervised learning
Signal use	combine realized returns	use current scores as features
Adaptation	fixed per anchor	model selected per anchor

Stage C: Expanding the Strategy Space

Objective 2: treat strategy scores as *features* for a supervised model.

Expand from 4 to **40 basis columns**:

$$A \in \mathbb{R}^{n \times 40} \quad \text{(top 10 parameter sets per strategy family)}$$

This mirrors the class model-selection exercise:

$$\text{many basis functions} \longrightarrow \text{select useful structure}$$

Four linear models compared:

OLS Ridge Lasso Elastic Net

Model selection criterion: **time-series cross-validated MSE** (no look-ahead).

Stage C: Forward Validation Design

Anchor-date setup

- Monthly anchor dates: 2020–2024
- **Selection period**: 1 year before anchor
  - fit model, pick best model by CV-MSE
- **Evaluation period**: after anchor
  - apply frozen model to unseen data
- Model re-fitted at each new anchor

$$\underbrace{\longleftarrow \text{1 year} \longrightarrow}_{\text{selection}} \Big|_{\text{anchor}} \underbrace{\longrightarrow}_{\text{evaluation}}$$

The model never sees future data during fitting.

This separates model discovery from forward validation.

Stage C: From Prediction to Portfolio Weight

The model outputs a predicted future cumulative return $\hat{y}_t$.

Convert to a long-only portfolio weight:

$$w_t = \text{clip}\left(\frac{\hat{y}_t}{s},\ 0,\ 1\right)$$

where s is a quantile-based scale factor from the selection period.

Execute via rolling tranches:

- Each day: compute signal, open one tranche at weight w_t
- Hold for fixed horizon (45 or 130 days), then close
- Gross exposure = average of all open tranches

Key property: when the model is confident ($\hat{y}_t$ large), exposure rises. When uncertain ($\hat{y}_t \approx 0$), exposure falls naturally.

Three-Stage Comparison: A vs B vs C

Hold 130 days · 1-month update · **Ret / Exposure / Efficiency (Ret ÷ Exp)**

Year	B&H	A: Single Signal	B: Ensemble	C: ML-Guided
2020	+23.9%	+4.4% / 27% / +16%	+6.4% / 32% / +20%	**+10.9%** / 57% / **+19%**
2021	−6.2%	+0.3% / 25% / +1%	+0.3% / 25% / +1%	**+0.8%** / 29% / **+3%**
2022	+0.8%	−0.9% / 46% / −2%	−2.3% / 42% / −5%	**−0.2%** / 31% / **−1%**
2023	+11.8%	+1.6% / 25% / +6%	+2.0% / 23% / +9%	**+4.0%** / 32% / **+12%**
2024	+27.0%	+5.9% / 21% / +29%	+8.0% / 29% / +27%	**+16.4%** / 65% / **+25%**

- C outperforms A and B in **every year** on both return and efficiency
- 2024: A(+5.9%) → B(+8.0%) → **C(+16.4%)** — clear step-up at each stage
- **Core insight**: ML-Guided learns *when* and *how much* to bet, not just which strategy

Stage C: What Does the Chosen Model Look Like?

Anchor: 2021-11-30 — OLS selected, horizon=130 days, corr=0.995

From 40 basis columns, the top contributing features (by absolute coefficient):

Feature	Strategy family	Coefficient
`ma_crossover_rank07`	MA Crossover	−0.212
`adaptive_band_rank05`	Adaptive Band	+0.099
`adaptive_band_rank06`	Adaptive Band	+0.097
`adaptive_band_rank03`	Adaptive Band	−0.075
`adaptive_band_rank04`	Adaptive Band	−0.073
`ma_crossover_rank10`	MA Crossover	+0.071
⋮	⋮	⋮

- 28 out of 40 features are nonzero — OLS uses the full correlated space
- The model mixes *ranks within a strategy*, not just strategies
- Prediction: $\hat{y}_t = \beta_0 + \sum_{j=1}^{40} \beta_j \cdot A_{tj}$
- $\hat{y}_t$ becomes the portfolio weight $w_t = \text{clip}(\hat{y}_t/s,\ 0,\ 1)$

How Often Should the Model Update?

Hold 130 days Ret / Avg. Exposure

Year	B&H	1M	2M	3M	6M
2020	+23.9%	+10.9% / 57%	+10.2% / 57%	+9.6% / 51%	+9.7% / 50%
2021	−6.2%	+0.8% / 29%	+0.3% / 35%	+0.6% / 28%	−0.7% / 49%
2022	+0.8%	−0.2% / 31%	+0.9% / 35%	−0.8% / 33%	−1.7% / 44%
2023	+11.8%	+4.0% / 32%	+4.1% / 23%	+3.8% / 40%	+3.4% / 15%
2024	+27.0%	**+16.4%** / 65%	+13.9% / 60%	+11.1% / 53%	+5.5% / 36%

Hold 45 days Ret / Avg. Exposure

Year	B&H	1M	2M	3M	6M
2020	+23.9%	+17.4% / 72%	+16.5% / 71%	+13.7% / 64%	+13.3% / 63%
2021	−6.2%	+1.5% / 35%	+1.8% / 37%	+1.2% / 34%	−1.2% / 49%
2022	+0.8%	−0.6% / 37%	+1.3% / 42%	−0.6% / 38%	−2.5% / 51%
2023	+11.8%	+2.5% / 44%	+3.7% / 34%	+0.9% / 50%	+3.8% / 24%
2024	+27.0%	**+25.2%** / 78%	+19.8% / 73%	+16.2% / 66%	+11.0% / 52%

Hold Horizon: 45 Days vs 130 Days

Stage C · 1-month update · **Ret** / **Exposure** / **Efficiency**

Year	B&H	Hold 130 days	Hold 45 days
2020	+23.9% / 100% / +24%	+10.9% / 57% / +19%	**+17.4%** / 72% / **+24%**
2021	−6.2% / 100% / −6%	**+0.8%** / 29% / +3%	+1.5% / 35% / **+4%**
2022	+0.8% / 100% / +1%	**−0.2%** / 31% / −1%	−0.6% / 37% / −2%
2023	+11.8% / 100% / +12%	**+4.0%** / 32% / +12%	+2.5% / 44% / +6%
2024	+27.0% / 100% / +27%	+16.4% / 65% / +25%	**+25.2%** / 78% / **+32%**

- **h45 captures bull markets better** (2020: +17.4%, 2024: +25.2%) — faster turnover
- **h130 more stable in flat/bear markets** (2022: −0.2% vs −0.6%, 2023: +4.0% vs +2.5%)
- **h45 efficiency in 2024 (+32%) exceeds B&H (+27%)** — same capital, better return per unit deployed
- Hold horizon is a hyperparameter: h45 for bull-market capture, h130 for stability

What's Next? — Priorities

1. **Multi-Asset Expansion** *(immediate — data already collected)*
   - Apply the same framework to BRK.B, oil, copper, equities
   - When GLD exposure is low, deploy idle capital into other assets
   - Cross-asset signals become additional columns of A
2. **Regime-Aware Exposure Control** *(short-term)*
   - Add a second-layer signal: *is the market currently predictable?*
   - Reduce exposure globally during high-uncertainty regimes
3. **Nonlinear Model Selection via SINDy** *(research)*
   - Expand the library with nonlinear terms: $\Theta(A) = [s_i,\ s_i^2,\ s_i s_j,\ \ldots]$
   - Sparse regression selects only the terms that matter
4. **Hierarchical Strategy Structure** *(research)*
   - Layer 1: direction signals $\longrightarrow$ Layer 2: timing signals
   - Each layer applies a nonlinear transform — this is a neural network

Conclusion

Main Finding

Treating trading strategy outputs as basis functions creates a valid, predictive strategy space. ML model selection over that space substantially outperforms both single-strategy and ensemble approaches.

Stage	Key Takeaway
A: Single Signal	Exposure 21–46%; captures small fraction of market move
B: Ensemble	Diversification alone does not improve prediction
C: ML-Guided	Learns *when* to be exposed; 2024: $\times 2.0$ vs B, $\times 2.8$ vs A

Live: Running on Raspberry Pi + Alpaca paper trading since April 2026.

Next:

- Multi-asset expansion (BRK.B, oil, copper) — data already collected
- Regime-aware exposure control layer
- Nonlinear model selection (SINDy)